MIL-L-8362C
10 Oct 1968
SUPERSEDING
MIL-L-8362B
15 April 1965

MILITARY SPECIFICATION

LUBRICANT, TIRE AND RIM, DEMOUNTING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid lubricant for tire and rim remounting.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250/5D	Aluminum Alloy Alclad 2024, Plate And Sheet
QQ-B-613	Brass, Leaded and Non-Leaded; Plate Rolled, Bar, Sheet and Strip
QQ-M-44	Magnesium Alloy Plate and Sheet
PPP-B-636	Box, Fiberboard
PPP-C-96	Cans, Metal, 28 Gage And Lighter

Military

MIL-E-7729	Enamel, Gloss
MIL-S-7809	Steel, Low Alloy, High Strength, Low Alloy, Bars, Shapes, Sheet, And Plates
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity

STANDARDS

Federal

FSC 2040

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FED-STD 595	Colors
FED-STD-601	Rudder Sampling And Testing
FED-STD- 791	Lubricants, Liquid Fuels And Related Products, Methods of Testing

Military

MIL-STD-105	Sampling Procedures And Tables For Inspectin by Attributes
MIL-STD-129	Marking For Shipment And Storage

2.2 Ocher publications.

DSAM 4145.3/AFM 71-4/TM 38-250/NAVWEPS 15-03-500/MCOP4030/19

Packaging And Handling Of Dangerous Material For Transportation By Military Aircraft

(Copies of specifications standards drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

American Trucking Associtations, Inc.

National Motor Freight Classification Rules

(Application for copies should be addressed co the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D.C. 20036).

Uniform Classification Committee

Uniform Freight CLassification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W Jackson Blvd., Chicago, Ill. 60606).

3. REQUIREMENTS

3.1 Preproduction This specification makes provisions for preproduction testing.

3.2 Material. Material used in the manufacture of the lubricant described herein shall be of good commercial quality and such as to produce a liquid lubricant suitable for the intended purpose and conforming to the requirements of this specification. The lubricant shall function as a tire demounting lubricant by facilitating the separtaion of the mounted tire bead from the wheel rim.

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3.3 Toxicity. Fumes of the lubricant shall not be injurious LO a person's health if inadvertently inhaled for a short period of time in a well ventilated area.

3.4 Homogeneity. The lubricant shall be homogeneous when examined as specified in 4.3.2.

3.5 Viscosity. The viscosity of the lubricant ac a temperature of 130 degrees ( ) Fahrenheit (F.) shall not vary by more than ±1 centistoke (cs) from the value obtained on the test specified in 4.3.3 after high and low temperature cycling.

3.6 Corrosiveness on metals. The lubricant shall not corrode metals when tested as specified in 4.3.4. There shall be no pitting, etching, or visible corrosion on the surface of any of the metals when viewed under a magnification of 20 diameters. Discoloration of erase or magnesium surfaces will be permitted.

3.7 Effect on protective finish. The lubricant shall have no deleterious effect on protective finishes when examined as specified in 4.3.5.

3.8 Deterioration of rubber. The lubricant shall not change the tensile strength, elongation, and volume of natural, Buns-S, and butyl unplasticized rubber specimens in excess of ±10 percent (%) of the orginal value when tested as specified in 4.3.6.

3.9 Coefficient of static friction. The coefficient of friction shall be not more than 0.3 when the Lubricant is tested as specified in 4.3.7 and shall be not less than 0.5 when tested as specified in 4.3.8.

3.10 Stability.

3.10.1 Storage stability No change in viscosity of more ban ±1 cs shall be permitted when samples are tested in accordance with 4.3.9.1, No changes in corrosive properties shall be permitted when samples are tested in accordance with 4.3.9.1.

3.10.2 Cycling stability. No change in viscosity of more than ±1 cs, and no change in corrosive properties shall be permitted after high temperature (+160°F.) and low temperature (-65°F.) cycling as specified in 4.3.9.2.

3.11 Product integrity Quality of the product shall be in accordance with good commercial pratice for this type of material.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein . Except as otherwise specified, the supplier may utilize his own facilities or any commercial

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laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests Inspection and testing of tire demounting lubricant shall be classified as follows

a. Preproduction tests. Preproduction tests are those tests required to determine conformance of the samples to the requirements specified herein. Preproduction tests consist of all tests in paragraphs 4.3.2 through 4.3.9.4 and the examination in 4.5.

D. Acceptance tests Acceptance tests are those tests accomplished on samples selected from each lot or batch produced for delivery under terms of the contract or purchase order.

4.3 Preproduction tests

4.3.1 Sampling Instructions. Preproduction test samples shall consist of two gallons of the tire denouncing lubricant. If the lubricant is packaged in concentrated form, the equivalent of the number of gallons specified herein shall be used. Samples shall be identified by tags marked with the following Information:

Sample for preproduction test
Lubricant, Tire and Rim, Demounting
Specification MIL-L-8362
Name of Manufacturer
Formulation, including percent composition by weight
Submitted by (name) (date) for preproduction testing in accordance with the requirements of MIL-L-8362 under contract or purchase order number (enter no.).

4.3.2 Examination of product The finished lubricant shall be examined to determine homogeneity. The lubricant shall be free from cakes, lumps, or hard gritty particles.

4,3.3 Viscosity Determination. The viscosity at 130°F. shall be determined in accordance with method 305.4 of FED STD 791.

4.3,4 Corrosion test. Cleaned and polished, 1 inch by 3 inch metal scrips of steel, brass, and magnesium shall be tested. The steel strip shall conform to MIL-S-7809 the brass strip shall conform to QQ-B-613 and the magnesium strip shall conform to QQ-M-44 The test shall De conducted by immersing the test strips, one each, in a 100 milliliter (ml), (20 centimeter(cm) by 3 cm), test tube containing 15 ml of lubricant test sample. The test tube shall be lightly stoppered with cork stoppers, supported in a beaker, and then placed in an oven at 1650F. for 72 hours. The strips shall be examined under magnification of 20 diameters every

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24 hours for evidence of corrosion.

4.3.5 Effect on protective finish test. Four test panels, 4 inches by 4 inches, shall be made from aluminum conforming to QQ-A-250/5b. Two test panels shall be prepared for testing by spraying with one coat of primer conforming to MIL-P-8585 followed by spraying with one coat of color No. 17038 (Glossy Black) enamel conforming to MIL-E-7729 type I. The two remaining test panels shall be prepared by spraying with one coat of primer conforming to MIL-P-8585 followed by spraying with enamel conforming to type I of MIL-E-7729, color to conform to color number 17038 of FED STD 595, The finished panels shall be dried for 46 hours. A few drops of the test lubricant shall be placed on the finished aluminum panels and allowed to stand at room temperature for approximately one week. At the end of this period the panels shall be thoroughly washed with water and examined for deleterious effects such as softening or lifting of the film.

4.3.6 Deterioration of rubber test. The deterioration of rubber test shall be conducted in accordance with FED STD 601 Methods 6111 and 6112. The unplasticized natural, SBR, and IIR rubber sheets shall conform to the composition data in Table I.

TABLE I

Formulation and Curing of Rubber Samples

	Butyl (IIR)		Buna-S (SBR)		Natural Rubber (NR) Smoked Sheet	
Formulation, Parts per 100	IIR	100	SBR	100	Smoked Sheet	100
	ELC Magnesia	3	Stearic Acid	1	Stearic Acid	3
	Zinc Oxide	5	Zinc Oxide	1	Zinc Oxide	5
	Methyl Tuads	1	Methyl Tuads	0.5	Sulphur	3
	Polyac	1	Sulphur	4	CAPTAX	1
	Sulphur	2	SRF BLACK	64	SRF BLACK	45
	Stearic Acid	2				
	SRF BLACK	61				
Curing Time[1]	20 minutes		20 minutes		20 minutes	
Temperature	310 degrees F		310 degrees F		310 degrees F	

[1] The rubber specimens are cured between sheets of aluminum foil in standard 6 inch by 6 inch by 0.070 inch ASTM moulds

4.3.7 Coefficient of static friction test, wet One test panel shall be prepared from unplasticized natural rubber sheet which conforms to the composition data in Table I. The test panel shall be 3 inch by 4 inch with a 1/8 inch diameter hole centered on a line 1/4 inch from one end of the panel. The panel shall be thoroughly cleaned with water and then rinsed with isopropyl alcohol. An additional test panel 6 inch by 6 inch by 0.91 inch of magnesium alloy conforming to QQ-M-44 shall be prepared. The magnesium panel shall be

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[illegible] lead[illegible] [illegible] [illegible] [

4.3.9 High temperature cycling test. Two, one pint, cardboard glass containers of the test samples shall be heated in an oven to 100°F. and maintained at that temperature for 24 hours. Then the sample shall be removed from the oven and cooled to room temperature for 24 hours, and again the sample shall be heated to 100°F for an additional 24 hours. The sample shall be cooled to room temperature. The viscosity and corrosion of the sample shall be determined.

4.4 Acceptance tests. If specified (see 6.2), acceptance tests shall be in accordance with the requirements specified herein. Acceptance tests shall consist of the examination in 4.3.3 and the tests specified in 4.3.4, 4.3.5, 4.3.6, 4.3.9, and 4.5.

4.4.1 Sampling for acceptance tests. Unless otherwise specified, a sample of not less than one gallon shall be selected from each lot or batch under inspection.

4.4.2 Rejection and retest. Failure of any lubricant sample to conform to one or more of the requirements of this specification shall result in rejection of the lot represented. Tire demounting lubricant which has been rejected may be reworked or replaced to correct the defects and retested for compliance to the requriements specified herein. before retesting full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the inspector. Material rejected after retest shall not be reworked or retested without the specific approval of the procuring activity

4.5 Examination of preparation for delivery An examination shall be made to determine that packaging, packing and marking requirements are complied with. Defects shall be scored in accordance with the list below. The lot size shall be the number of shipping containers prepared for delivery with the exception that containers need not be sealed or closed . Examination shall be made in two phases, first an interior examination and second an examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be one container prepared for delivery as set forth above , The inspection level shall be S-2 of MIL-STD-105, with an acceptable quality level (AQL) of 4.0 defects per 100 units.

EXAMINE	DEFECTS
Marking (interior and e interior containers)	Oml.ted, incorrect, illegible , improper size, sequence, or method of application
Materials	Components missing. damaged, defective or not as specified
Workmanship	Inadequate or improper closure of cans, damaged or defective cans; im-

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	proper positioning of cans or improper protection of bridge handles, not snugly fitted in exterior containers; fiberboard pad, top or bottom omitted; improper closure of exterior container; loose strapping, bulging or distortion of containers.
Contents (exterior container)	Number per container not as specified.

5 PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be Level A or C as specified (see 6.2)

5.1.1 Level A. Lubricant tire and rim, demounting shall be packaged in Type V, oval or oblong, Class 5, 1 gallon cans with bridge handles, conforming to PPP-C-96. Closure of cans shall be in accordance with the appendix to PPP-C-96.

5.1.2 Level C. Lubricant shall be packaged in accordance with the suppliers commercial practice in the number of gallons per unit container specified (see 6.2) .

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2). Shipping containers, insofar as practical, shall be uniform an size and shape and of minimum cube and tare weight consistent with the protection required, and shall contain identical quantities. Shipments consigned to a single consignee shall be unitized loaded and secures on pallets with four way fork lift entry to facilitate handling and scoring LO the maximom extent consistent with economy The lubricant shall be prepared for shipment in accordance with CAB, ICC, Sate or Postal regulation or Military regulation referenced in 5.3.3 as applicable LO the mode of transportation for shipment of dangerous material.

5.2.1 Level A. Lubricant unit packaged in accordance with paragraph 5.1.1 shall be packed 6 each (1 gallon cans) in a box. fiberboard. conforming to Class 2, V30, Style RCS, PPF-B-636 The for shall be of proper size to afford a snug fit for a 2 x 3 single tier stacking arrangement with can closures placed toward the outside of the box The bridge handle shall be protected as specified in the appendix to PPP-C-96 The fiberboard box shall have one (1) peice fiberboard pads on top and bottom of its concents Material for pads shall be the same as for the box.

5.2.2 Level B Same as 5 2 1 except box, fiberboard and pads shall conform to Class 1, domestic service fiberboard

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5.2. 3 Level C. Lubricant packaged in accordance with 5.1, which requires over-packing for acceptance by the carriers shall be packed in exterior shipping containers in a manner that will ensure protection against damage from transporta tion and handlin g at the lowest rate to the point of de. livery , Containers shall comply with uniform freight classification rules or National Motor Freight Classification Rules.

5.3 Marking.

5.3.1 Except as otherwise specified (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129

5.3.2 Unless otherwise specified (see 6 2), the method of marking and the caution label for terneplate cnas specified in section 6 of PPP-C-96 shall apply.

5.3.3 Shipment by military aircraft. When the properties of any lubricant consist of flammable or poisionous liquids or an oxidizing agent, and a mode of transportation to be utilized is military aircraft the containers shall be marked in accordance with DSAM 4145.3/AFM 71-4/TM38-250/NAVWEPS 15-06-500/MCOB 4030/194

6. NOTES .

6.1 Intended use. The lubricant is intended for use in effecting easier tire dismounting by lubrication the interface of the tire bead and the tire rim and in effecting more satisfactory bead, tube and flap positioning during mounting of tires

6.2 Ordering data. Procure,emt documents should specify the following:

a. Title, number and date of this specification.

o. Quantity of lubricant desired.

c. Acceptance tests, if required (see 4.4).

d. Levels of preservation and packaging required (see 5.1).

e. Size of unit container for Level C packaging (see 5.1.2) ,

f. Level of packing required (see 5.2).

g. Marking if different (see 5.3.1 and 5.3.2).

6 3 Preproduction samples Preproduction samples may be selected from quantiti es of lubricant produced for preproduction tests or may be selected from regular production In either case, he samples shall be identical with the proposed product and shall have been made by the methods that are to be used or were used in production. of items to fulfill the terms

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of the contract or purchase order. Approval of preproduction st shall not relieve the supplier of the responsibility to furnish duct in accordance with the requirements of this specification.

6.4 Batch. A batch is defined as that quantity of material whi been manufactured by some unit chemical process and subjected to physical mixing operation intended to sake the final product suc uniform.

6.5 Asterisks are not used in this revision to identify changes respect to the previous issue, due to the extensiveness of the c

Custodians:

Army-- AT
Navy - YD (Code 310)
Air Force - 84

Preparing Activity:

Air Force - 84

Project Number,
2640-0071

Reviewer Activities:

Army - AV
Navy - YD - UP

Code "N"

User Activities:

Navy - MC
Army-ME

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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